

MANDURAH, Australia

WMO: 946050

Lat: 32.5219S

Lon: 115.7119E

Elev: 11

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	44.2	46.4	34.8	29.6	66.3	37.0	32.3	60.9	32.1	61.7	29.1	61.5	4.5	110	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.9	94.4	67.4	90.9	67.3	87.8	66.6	72.4	82.9	71.0	81.9	69.8	80.4	9.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.8	109.7	75.4	68.1	103.3	74.4	66.5	97.6	74.0	36.2	82.2	34.9	82.2	33.9	80.0	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.6	22.1	19.2	DB	41.5	101.3	2.7	2.8	39.6	103.3	38.0	104.9	36.5	106.5	34.6	108.5	(4)
(5)			WB	39.4	74.8	2.0	1.7	37.9	76.0	36.7	77.0	35.6	77.9	34.1	79.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.4	74.4	74.9	72.4	67.5	62.5	58.6	56.9	57.6	59.0	62.7	67.7	71.5	(6)	
	DBStd	7.95	5.50	5.71	5.41	4.58	3.86	3.45	3.49	3.29	3.69	4.42	5.15	6.09	(7)	
	HDD50	1	0	0	0	0	0	0	1	0	0	0	0	0	(8)	
	HDD65	1112	2	1	5	24	96	192	250	228	182	99	26	7	(9)	
	CDD50	5637	757	697	696	525	387	259	216	237	271	394	532	666	(10)	
	CDD65	1270	293	278	236	99	18	0	0	0	3	28	107	208	(11)	
	CDH74	9886	2594	2464	1771	435	32	0	0	1	14	168	710	1697	(12)	
	CDH80	3641	1033	989	603	85	1	0	0	0	34	219	677		(13)	
(14) Wind	WSAvg	9.6	10.7	10.2	9.4	8.1	8.2	8.9	9.6	9.5	9.9	9.6	10.7	10.7	(14)	
(15) Precipitation	PrecAvg	33.1	0.4	0.6	0.8	1.7	4.6	6.9	6.9	4.6	3.2	1.7	1.1	0.4	(15)	
	PrecMax	42.3	3.1	4.3	4.1	4.3	10.2	13.4	12.4	7.5	6.1	4.9	5.0	2.1	(16)	
	PrecMin	20.7	0.0	0.0	0.0	0.0	1.3	2.6	3.4	1.4	0.9	0.0	0.2	0.0	(17)	
	PrecStd	5.0	0.7	0.9	1.0	1.3	2.0	2.6	2.0	1.4	1.4	1.0	0.9	0.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.3	97.7	95.3	87.0	77.3	71.1	68.0	70.5	75.8	84.6	91.8	97.8	(19)	
		MCWB	68.4	69.3	68.3	63.8	61.3	58.7	58.7	58.6	59.9	63.0	65.0	68.0	(20)	
	2%	DB	93.4	93.4	89.9	82.2	74.0	68.1	65.6	67.1	70.7	78.8	86.2	92.2	(21)	
		MCWB	67.2	67.9	67.1	64.2	61.8	58.8	58.6	59.3	59.3	61.5	63.8	66.8	(22)	
	5%	DB	89.2	89.7	86.3	78.9	71.7	66.5	64.4	65.5	67.9	74.4	81.6	87.5	(23)	
		MCWB	67.6	67.8	66.0	64.1	61.2	58.8	57.9	58.5	59.3	60.7	63.1	65.8	(24)	
	10%	DB	85.8	86.3	82.9	75.7	69.7	65.2	63.1	64.0	65.7	70.9	77.7	83.2	(25)	
		MCWB	66.9	67.2	65.3	63.2	61.3	58.3	57.0	57.6	58.4	60.3	62.7	65.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.7	75.4	72.8	70.8	68.1	64.6	63.1	64.7	64.8	66.5	69.2	71.9	(27)	
		MCDB	82.6	85.6	80.5	74.1	70.4	66.5	64.9	66.3	68.3	73.4	77.1	85.2	(28)	
	2%	WB	72.4	72.5	71.1	69.0	66.2	62.7	61.1	62.1	62.9	65.0	67.5	70.1	(29)	
		MCDB	83.6	83.9	79.9	73.9	69.7	65.5	63.6	64.4	66.6	72.0	76.6	83.2	(30)	
	5%	WB	70.6	70.8	69.8	67.4	64.5	61.0	59.6	60.3	61.2	63.6	66.2	68.6	(31)	
		MCDB	82.3	82.9	78.8	74.0	68.9	64.4	62.7	63.7	65.8	70.2	75.7	81.3	(32)	
	10%	WB	69.1	69.5	68.4	65.7	62.7	59.6	58.2	58.8	59.5	62.1	64.8	67.1	(33)	
		MCDB	80.9	81.6	78.5	73.1	68.0	63.5	61.9	62.7	64.7	68.8	74.7	79.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	17.9	17.0	14.9	13.6	12.3	11.4	11.8	12.6	14.6	15.4	16.6	(35)	
		MCDBR	24.9	23.8	22.4	19.4	16.6	13.8	11.9	14.0	16.7	21.2	22.9	24.5	(36)	
	5% WB	MCWBR	10.0	9.7	9.1	9.3	9.7	8.9	8.3	9.1	9.9	10.6	10.6	10.1	(37)	
		MCWBR	17.5	17.3	14.7	12.7	11.5	10.7	9.6	10.4	12.6	14.8	15.9	19.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.337	0.336	0.324	0.316	0.310	0.305	0.303	0.311	0.331	0.327	0.329	0.328	(40)	
		taud	2.599	2.614	2.646	2.656	2.641	2.634	2.639	2.609	2.526	2.556	2.575	2.599	(41)	
	Ebn at Noon		318	312	305	291	276	270	276	288	298	311	318	322	(42)	
		Edh at Noon	33	31	28	25	23	22	23	26	31	33	33	33	(43)	
(44) All-Sky Solar Radiation	RadAvg		2580	2319	1903	1365	1006	808	858	1136	1529	2014	2419	2641	(44)	
	RadStd		121	87	92	84	64	55	57	62	94	84	114	105	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)		N/A	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page